The documentation and process conversion measures necessary to comply with this revision shall be completed by 3 October 2000

INCH-POUND

MIL-PRF-19500/427G
3 April 2000
SUPERSEDING
MIL-PRF-19500/427F
15 January 1998

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
TYPES 1N5614, 1N5616, 1N5618, 1N5620, 1N5622, 1N5614EG1, 1N5616EG1, 1N5618EG1, 1N5620EG1, 1N5622EG1, 1N5614UL, 1N5616UL, 1N5618UL, 1N5620UL, 1N5622UL, 1N5614US, 1N5616US, 1N5618US, 1N5620US, 1N5622US, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for silicon, hermetically sealed power rectifier diodes. Four levels of product assurance are provided for each encapsulated device as specified in MIL-PRF-19500. Two levels of product assurance are provided each unencapsulated device type.

1.2 Physical dimensions. See figures 1, (similar to DO-204; formerly DO-41), 2, 3, 4, and 5 (JANHC and JANKC).

* 1.3 Maximum ratings.

Types	V_R	V_{RWM}	I_O		I_{FSM} T_A = +100°C I_0 = 750 mA dc t_p = 8.3 ms		T_{STG} and T_J	t_{rr}	$R_{\theta JL}$ 4/
			T_A = +55°C 1/ 2/	T_A = +100°C 2/ 3/		EG1 suffix			
	V dc	V(pk)	A dc	mA dc	A(pk)	A(pk)	°C	µs	See figure 6
1N5614, UL, US, EG1	200	200	1	750	30	25	-65 to +175	2	
1N5616, UL, US, EG1 5/	400	400	1	750	30	25	-65 to +175	2	
1N5618, UL, US, EG1 5/	600	600	1	750	30	25	-65 to +175	2	
1N5620, UL, US, EG1 5/	800	800	1	750	30	25	-65 to +175	2	
1N5622, UL, US, EG1 5/	1,000	1,000	1	750	30	25	-65 to +175	2	

1/ From 1 A dc at T_A = +55°C, to 0.75 A dc at T_A = +100°C, derate linearly at 5.56 mA/°C.
2/ For the 1 A dc rating at the +55°C and the 750 mA dc rating at +100°C, no special mounting, heat sinking, or forced air flow across exposed areas of the device is required.
3/ From 0.75 A dc at T_A = +100°C, to 0 A dc at T_A = +175°C, derate linearly at 10 mA/°C.
4/ $R_{\theta JL} \leq 38$°C/W at L = .375 inch (9.53 mm) for UL and US suffix type devices, $R_{\theta JEC}$ = 7.0°C/W.
5/ Barometric pressure reduced: 1N5616 and 1N5618 = 8 mm Hg, 1N5620 and 1N5622 = 33 mm Hg (not for UL and US suffix type devices).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, Post Office Box 3990, Columbus, OH 43213-1999, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

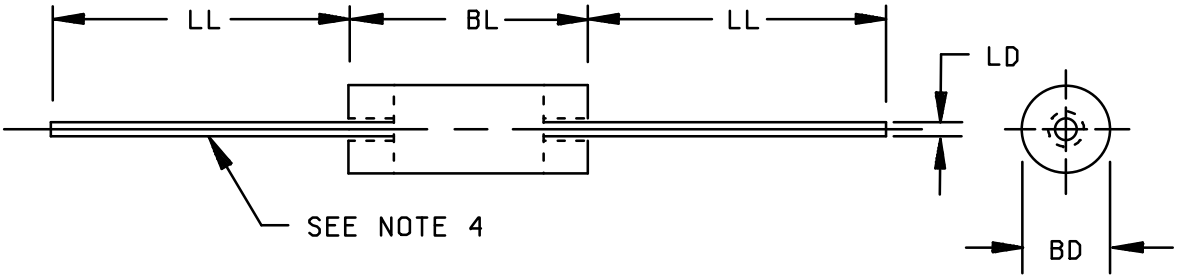
3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

EG	Epoxy over Glass hermetically sealed EG1 case.
UL	Long-body unleaded or surface mounted diodes (square end-caps).
US	Short-body unleaded or surface mounted diodes (square end-caps).

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and in figures 1, 2, 3, 4, and 5 herein. Except for JANHC and JANKC, all devices shall be metallurgically bonded-thermally-matched-noncavity-double plug construction, utilizing a category I bond, in accordance with MIL-PRF-19500. UL and US version devices shall be structurally identical to the nonsurface mount version devices except for lead configuration.

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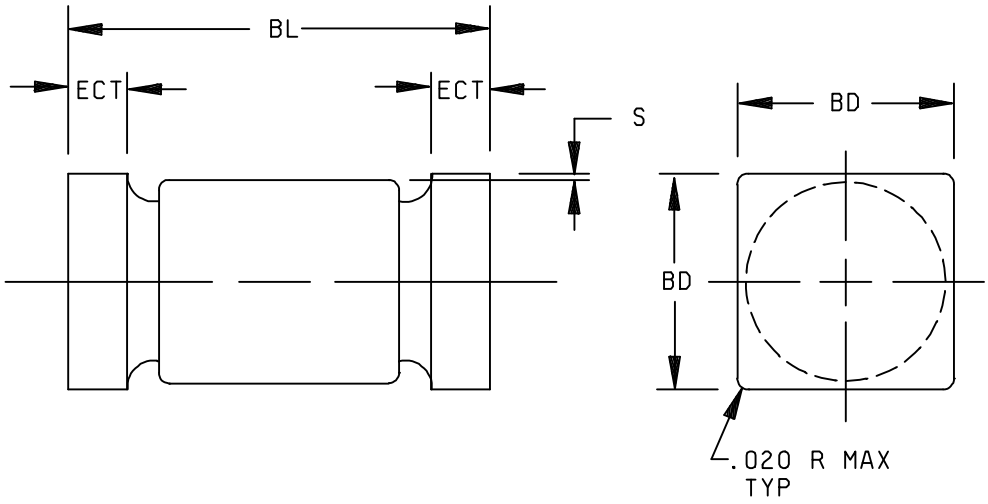


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	0.065	0.110	1.65	2.79	3
BL	0.130	0.225	3.30	5.72	4
LD	0.026	0.033	0.66	0.84	
LL	1.00	1.30	25.4	33.02	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension BD shall be measured at the largest diameter.
4. The BL dimension shall include all uncontrolled areas of the device.
5. The shape of the body, within the bounds of the dimensions is optional.

FIGURE 1. Physical dimensions for axial leaded devices only (similar to DO-214, formerly DO-41).

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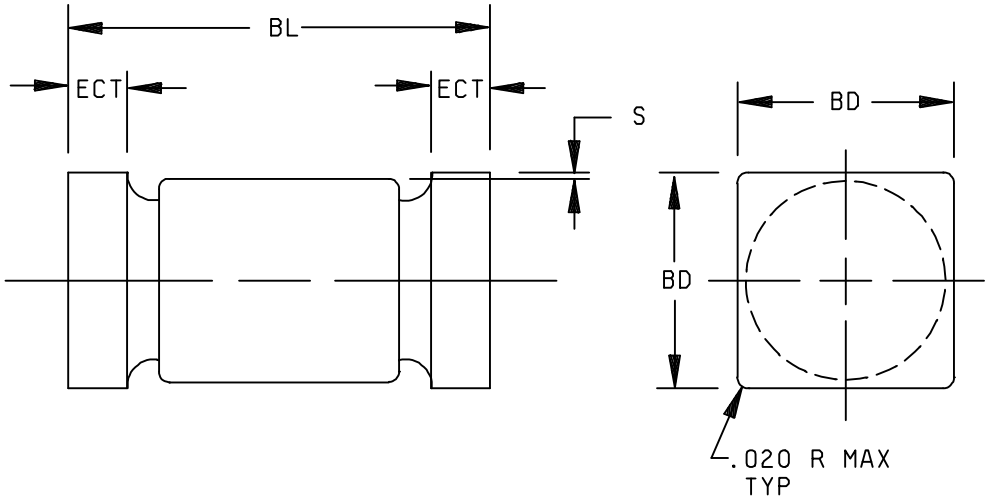


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.091	.125	2.31	3.18
BL	.168	.225	4.27	5.72
ECT	.019	.028	0.48	0.71
S	.003	- - -	0.08	- - -

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions types 1N5614UL, 1N5616UL, 1N5618UL, 1N5620UL, and 1N5622UL (surface mount devices).

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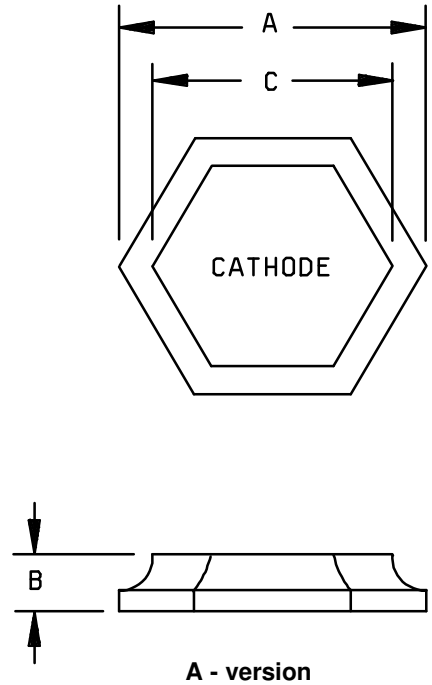


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.091	.103	2.31	2.62
BL	.168	.200	4.27	5.08
ECT	.019	.028	0.48	0.71
S	.003	- - -	0.08	- - -

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Physical dimensions types 1N5614US, 1N5616US, 1N5618US, 1N5620US, and 1N5622US (surface mount devices).

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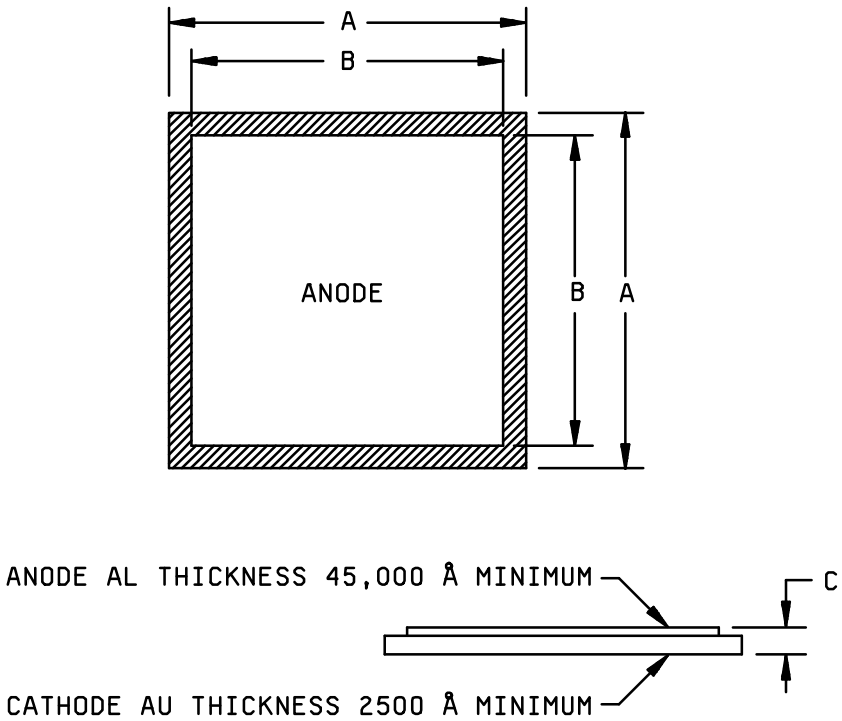
A - version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.047	.053	1.19	1.35
B	.007	.011	0.18	0.28
C	.033	.037	0.84	0.94

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Top metal: Gold 10,000 Å minimum.
   Back metal: Gold 4,000 Å minimum.

FIGURE 4. Physical dimension, JANHCA AND JANKCA die.

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B - version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.039	.043	1.00	1.09
B	.021	.025	0.53	0.64
C	.008	.012	0.20	0.31

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Top metal: Aluminum (anode) . . . . . 45,000 Å minimum.
   Back metal: Gold (cathode) . . . . . 2,500 Å minimum.

FIGURE 5. Physical dimension, JANHCB AND JANKCB die.

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3.5 Marking of UL and US version devices. For UL and US version devices only, all marking (except as stated in 3.6) may be omitted from the body, but shall be retained on the initial container.

3.6 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for UL and US suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end or a contrasting colorband may be used.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in Table I, Group A herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 JANHC and JANKC devices. Qualification for die shall be in accordance with appendix H of MIL-PRF-19500.

* 4.3 Screening (JANTX, JANTXV and JANS levels only). For appendix D qualified suppliers, screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable. For appendix C qualified suppliers (QML), refer to QM plan for screening requirements.

Screen (see appendix E, table IV of MIL-PRF -19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	I_{R1} and V_F	Not applicable
11	I_{R1} and V_F, $\Delta I_{R1} \leq$ 100 percent of initial reading or ± 100 nA dc, whichever is greater. $\Delta V_F \leq \pm$ 0.1 V dc.	I_{R1} and V_F
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein: $\Delta I_{R1} \leq$ 100 percent of initial reading or ± 100 nA dc, whichever is greater. $\Delta V_F \leq \pm$ 0.1 V dc. Scope display evaluation (see 4.5.3)	Subgroup 2 of table I herein: $\Delta I_{R1} \leq$ 100 percent of initial reading or ± 100 nA dc, whichever is greater. $\Delta V_F \leq \pm$ 0.1 V dc. Scope display evaluation (see 4.5.3)

1/ Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.

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4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038, condition B of MIL-STD-750, T_A = room ambient as designated in 4.5 of MIL-STD-750, V_R = full rated V_{RWM} (see 1.3); f = 50-60 Hz, I_O = 1 A dc, T_{EC} = +100°C to +125°C for surface mount devices (UL and US version) (see 4.5.7).

4.3.2 Screening (JANHC or JANKC). Screening of die shall be in accordance with MIL-PRF-19500, appendix G.

4.3.3 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. Maximum limit = 4.5°C/W.

4.3.3.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record value $Z_{\theta JX}$) and shall be supplied to the qualifying activity on a 500 piece sample from the qualification lot prior to qualification approval.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$ in a group A, subgroup 2 inspection:

a. I_M measuring current..... 10 mA.

b. I_H forward heating current ..... 10 A.

c. t_H heating time..... 10 ms.

d. t_{MD} measurement delay time ..... 100 µs maximum.

The maximum limit for $Z_{\theta JX}$ in group A, subgroup 2 is $Z_{\theta JX}$ (maximum) = 4.5°C/W.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein. Delta measurements shall be as specified in table II herein. $Z_{\theta JX}$ need not to be performed. Leaded samples may be used in lieu of surface mount devices (UL and US version) for life tests.

* 4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	4066	I_{FSM} = rated (see 1.3); 10 surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 750 mA dc; V_{RWM} = rated V_{RWM} (see 1.3); T_A = +100°C.
B4	1037	I_O = 1 A dc minimum; T_A = Room Ambient; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3); t_{on} = t_{off} = 3 minutes minimum, 2,000 cycles.

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B5	1027	I_O = 1 A dc minimum; V_R = rated V_{RWM} (see 1.3); T_A = +150°C minimum, adjust I_F and T_A as required to give an average T_J = +275°C +0°, -35°C (see 4.5.7).
B6	3101 or 4081	+25°C ≤ T_A ≤ +35°C (recorded before test is performed); $R_{\theta JL}$ (maximum) ≤ 38°C/W; L = .375 inch (9.53 mm) (see figure 6). For surface mount devices (UL and US version), $R_{\theta JEC}$ ≤ 7°C/W.
B7	- - -	Peak reverse power, see 4.5.4 and figure 7 herein. P_{RM} ≥ 500 W. Test shall be performed on each sublot; sampling plan = 10, end-points, see 4.4.2.

* 4.4.2.2 Group B inspection, appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	4066	I_{FSM} = rated (see 1.3); 10 surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 750 mA dc; V_R = rated V_{RWM} (see 1.3); T_A = +100°C.
B3	1027	T_A = Room Ambient, I_O = 1 A dc minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3, 4.5.7, and 4.5.7.1).
B5	3101 or 4081	+25°C ≤ T_A ≤ +35°C (recorded before test is performed); $R_{\theta JL}$ (maximum) ≤ 38°C/W; L = .375 inch (9.53 mm) (see figure 6). For surface mount devices (UL and US version), $R_{\theta JEC}$ ≤ 7°C/W.
B7	- - -	Peak reverse power, see 4.5.4 and figure 7 herein. P_{RM} ≥ 500 W. Test shall be performed on each sublot; sampling plan = 10, end-points, see 4.4.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein. Delta measurements shall be as specified in table II herein. $Z_{\theta JX}$ need not to be performed. Leaded samples may be used in lieu of surface mount devices (UL and US version) for life tests.

* 4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Tension: Test condition A, weight = 12 pounds. Lead fatigue: Test condition E, weight = 1 pound. NOTE: Neither tension nor lead fatigue are applicable to surface mount devices (UL and US version).
C6	1026	T_A = Room Ambient, I_O = 1 A dc minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3, 4.5.7, and 4.5.7.1).

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500.

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4.4.4.1 Group E inspection, appendix E, table IX of MIL-PRF-19500.

Subgroup	Method	Conditions
E5	1001	1N5616, 1N5618 = 8 mm Hg (100,000 ft); 1N5620, 1N5622 = 33 mm Hg (70,000 ft); $V_R = V_{RWM}$ (see 1.3); t = 1 minute (minimum).

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

* 4.5.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature; T_A = Room Ambient.

* 4.5.3 Scope display evaluation. Scope display evaluation shall be in acccordance with MIL-STD-750 method 4023. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 5 to 20 µA/division and 50 to 200 V/division. Reverse current over the knee shall be at least 50 µA. Each device may exhibit a sharp knee characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.

4.5.4 Peak reverse power test. This test shall be measured in the circuit of figure 7, or equivalent. A 20 microsecond half-sine waveform of current shall be used and peak reverse power shall be determined by the product of peak reverse voltage and peak reverse current.

4.5.6 Thermal resistance. Thermal resistance measurement shall be performed in accordance with MIL-STD-750, method 3101 or 4081. Forced moving air or draft shall not be permitted across the device during test. The maximum limit for $R_{\theta JL}$ under these test conditions shall be $R_{\theta JL(max)}$ = 38°C/W at .375 inch (9.53 mm) and 7°C/W at 0 inch (0.00 mm).

4.5.7 Life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180° nor less than 150°.

4.5.7.1 Mounting conditions. At the option of the manufacturer, any clips or heat sink mounting configurations may be utilized provided that I_o is adjusted such that the junction temperature of each diode is maintained at least +120°C above the specified ambient temperature of +25°C.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions 1/		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Forward voltage	4011	I_F = 3 A dc, pulsed (see 4.5.1) t_p = 300 µs, 2 maximum duty cycle	V_F	0.8	1.3	V dc
Reverse current leakage	4016	DC method; V_R = rated V_R (see 1.3)	I_{R1}		0.5	µA dc
Breakdown voltage	4021	I_R = 50 µA dc	$V_{(BR)R}$			
1N5614, UL, US, EG1				200		V dc
1N5616, UL, US, EG1				400		V dc
1N5618, UL, US, EG1				600		V dc
1N5620, UL, US, EG1				800		V dc
1N5622, UL, US, EG1				1,000		V dc
Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		4.5	°C/W
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	DC method; V_R = rated V_R (see 1.3)	I_{R2}		25	µA dc
Subgroup 4						
Reverse recovery time	4031	Condition B1	t_{rr}		2	µs
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups B, and C delta measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Reverse current leakage change	4016	DC method	ΔI_{R1} 4/		for JANTX and JANTXV, ≤ 250 nA dc or 100 percent, whichever is greater; for JANS, ≤ 100 nA dc or 100 percent, whichever is greater.	
2.	Forward voltage change	4011	I_F = 3 A dc; pulsed (see 4.5.1)	ΔV_{F1} 4/		± 50 mV dc maximum change from previous measured value.	

1/ The electrical measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table II herein, step 2.

b. Subgroup 4, see table II herein, step 2.

c. Subgroup 5, see table II herein, steps 1 and 2.

2/ The electrical measurements for appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table II herein, step 1.

b. Subgroup 6, see table II herein, step 1.

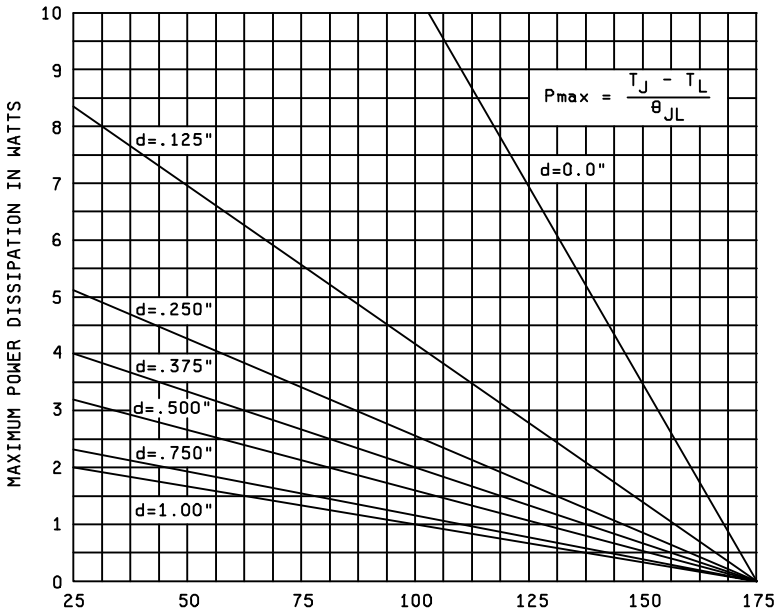
3/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2 (JANS).

b. Subgroup 6, see table II herein, steps 1 and 2 (JANS), step 1 (JANTX and JANTXV).

4/ Devices which exceed the group A limits for this test shall not be accepted.

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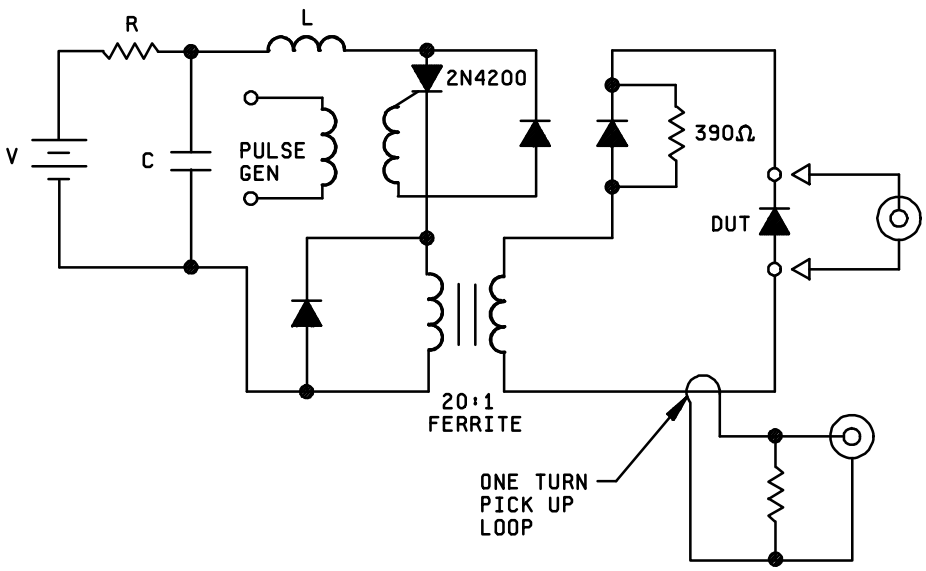
Maximum lead temperature in °C (T_L at point d from body (see table below)).
For maximum operating junction temperature of +175° with equal two-lead conditions.

d (see notes 1 and 2)	$R_{\theta JL}$ (see note 3)
Dimensions	°C/W
0.000 (0.00 mm)	7
0.125 (3.18 mm)	18
0.250 (6.35 mm)	29
0.375 (9.53 mm)	38
0.50 (12.7 mm)	45
0.75 (19.1 mm)	62
1.00 (25.4 mm)	76

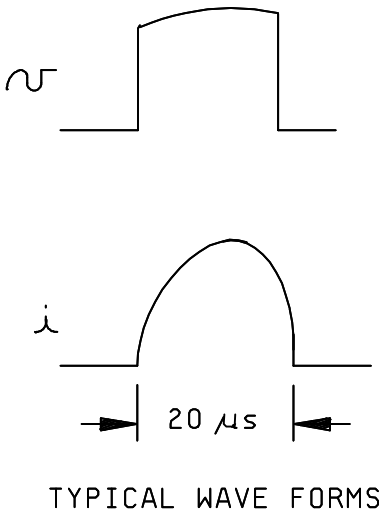
1. Lead distance point d and thermal resistance data.
2. Thermal resistance test performed at .375 inches (9.53 mm).
3. For US suffix devices: $R_{\theta JEC}$ = 7.0°C/W.

FIGURE 6. Power derating curves maximum power in watts versus lead temperature.

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NOTES:
1. L - 13T #22 pm 1 inch (25.4 mm) diameter form (air core).
2. C - 1 to 10µfd to give 20 µs pulse width.
3. V - adjustable to 200 volts for power desired in device under test.



TYPICAL WAVE FORMS

FIGURE 7. Peak reverse power measurement circuit and waveforms.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5614) will be identified on the Qualified Manufacturers List (QML).

JANHC and JANKC ordering information		
PIN	Manufacturer	
	14552	33178
1N5614	JANHCA1N5614 JANKCA1N5614	JANHCB1N5614 JANKCB1N5614
1N5616	JANHCA1N5616 JANKCA1N5616	JANHCB1N5616 JANKCB1N5616
1N5618	JANHCA1N5618 JANKCA1N5618	JANHCB1N5618 JANKCB1N5618
1N5620	JANHCA1N5620 JANKCA1N5620	JANHCB1N5620 JANKCB1N5620
1N5622	JANHCA1N5622 JANKCA1N5622	JANHCB1N5622 JANKCB1N5622

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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - CR
Navy - EC
Air Force - 11
NASA – NA
DLA - CC

Review activities:
Army - AR, MI, SM
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Preparing activity:
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3. **DOCUMENT TITLE** SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, TYPES 1N5614, 1N5616, 1N5618, 1N5620, 1N5622, 1N5614EG1, 1N5616EG1, 1N5618EG1, 1N5620EG1, 1N5622EG1, 1N5614UL, 1N5616UL, 1N5618UL, 1N5620UL, 1N5622UL, 1N5614US, 1N5616US, 1N5618US, 1N5620US, 1N5622US, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

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5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43213-1999	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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